MILITARY SPECIFICATION

AMPLIFIERS, RESOLVER, GENERAL SPECIFICATION FOR

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the general requirements and test procedures for non-repairable (incapsulated) transistorized resolver amplifiers.

1.2 Classification - Amplifiers covered by this specification shall be in accordance with the requirements of the applicable specification sheets.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-3965	Capacitors, Fixed, Electrolytic, Tantalum
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

FSC 5990

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Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment (ASG)
MIL-I-16910	Interference Measurement, Radio, Methods and Limits, 14 Kilocycles to 1000 Megacycles
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-B-22205	Bags, Transparent, Flexible, Heat Sealable for Packaging Applications
MIL-R-22973	Reliability Index Determination for Avionic Equipment Models, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for shipment and storage
MIL-STD-756	Reliability of Weapons Systems, Procedures for Prediction and Reporting Prediction of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

(When requesting any of the applicable documents, refer to both title and number. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific, procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets - The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Preproduction - The resolver amplifiers furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction tests specified herein.

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3.3 Quality conformance - Each amplifier furnished under this specification shall have demonstrated that it fulfills all of the requirements of this specification by satisfactorily performing all of the individual tests specified herein, and being one of a lot from which representative units have passed the specified Sampling tests.

3.4 Design and construction -

3.4.1 Compliance with Specification MIL-E-5400 - Amplifiers furnished under this specification shall comply with all requirements of MIL-E-5400 with the following exceptions:

a. Requirements in conflict with the applicable specification sheets.

b. Requirements in conflict with this specification.

c. Requirements which are not applicable, such as those pertaining to materials, parts, or processes not used in the construction of the amplifier.

3.4.2 Adjustments - Unless otherwise specified in the applicable specification sheets, no adjustment for gain or phase shift shall be provided.

3.4.3 Capacitors - Capacitors used throughout the resolver amplifier shall conform to MIL-C-3965.

3.4.4 Semiconductors - Silicon semiconductors conforming to MIL-S-19500 shall be used throughout the resolver amplifier.

3.4.5 Non-magnetic materials - Non-magnetic materials shall be used for all parts of the resolver amplfier except where magnetic properties are essential.

3.4.6 Magnesium - Magnesium or magnesium alloys in any form shall not be used.

3.4.7 Physical characteristics - The physical characteristics of the amplifier shall be in accordance with the applicable specification sheets.

3.4.8 Electrical characteristics -

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3.4.8.1 Signal input voltage - The range of the signal input voltage shall be specified in the applicable specification sheets.

3.4.8.2 High transient signal effect - The high transient signal effect test shall be performed in accordance with paragraph 4.3.4. It shall show no damage and meet the requirements of this specification.

3.4.8.3 Maximum DC Power supply current - When the amplifier is tested in accordance with paragraph 4.3.5, the DC Power supply current shall not exceed the value specified in the applicable specification sheets.

3.4.8.4 Power input - The power input to the amplifier when tested in accordance with paragraph 4.3.6, shall be specified in the applicable specification sheets.

3.4.8.5 Phase shift - When the amplifier is tested in accordance with 4.3.7, the phase shift shall be as sepcified in the applicable specification sheets.

3.4.8.6 Input impedance - When the amplifier is tested in accordance with 4.3.8, the input impedance shall be as specified in the applicable specification sheets.

3.4.8.7 Output impedance - When the amplifier is tested in accordance with 4.3.9, the output impedance shall be as specified in the applicable specification sheets.

3.4.8.8 Temperature and altitude - The results of the temperature and altitude tests obtained from 4.3.10 shall be as specified in the applicable specification sheets.

3.4.8.9 Internal noise - When the amplifier is tested in accordance with 4.3.11, the internal noise level shall not exceed the value specified in the applicable specification sheets.

3.4.8.10 Warm up - When the resolver amplifier is tested in accordance with 4.3.12, the warm up time shall not exceed the value specified in the applicable specification sheets.

3.4.8.11 Voltage gain and linearity - When the amplifier is tested in accordance with 4.3.13, the results shall be as specified in the applicable specification sheets.

3.4.8.11.1 Resonant frequencies (closed loop) - There shall be no resonant frequencies except the ones specified in the applicable specification sheets when the resolver amplifier is tested in accordance with 4.3.13.d.

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3.4.8.12 Vibration - The amplifier shall be subjected to the vibration test specified in paragraph 4.3.14.

3.4.8.13 Shock - The amplifier shall be subjected to the shock test specified in paragraph 4.3.15.

3.4.8.14 Humidity - The amplifier shall be subjected to the humidity test specified in paragraph 4.3.16.

3.4.8.15 Salt spray - The amplifier shall be subjected to the salt spray test specified in paragraph 4.3.17.

3.4.8.16 Sand and dust - The amplifier shall be subjected to the sand and dust test specified in paragraph 4.3.18.

3.4.8.17 Fungus - The amplifier shall be subjected to the fungus test specified in paragraph 4.3.19.

3.4.8.18 Acceleration - The resolver amplifier shall be tested as specified in 4.3.20 and in accordance with Procedure I of MIL-E-5272.

3.4.8.19 Radio interference - The amplifier shall be so constructed that radiated and conducted radio noise will not interfere with the effective use of communication or navigation equipment. (MIL-I-16910).

3.4.8.20 Reliability - The resolver amplifier reliability shall be planned, designed, analyzed, and reported in accordance with MIL-STD-756. The reliability goal, expressed in mean-time-between-failure (MTBF), shall be as specified in the applicable specification sheets, at a confidence level of 90 percent. To determine the MTBF actually attained, the resolver amplifier shall be tested in accordance with MIL-R-22973 according to the level specified in the applicable specification sheets. The quantity of resolver amplifiers to be tested shall be as specified in the contract. For preproduction the test duration and the maximum number of failures shall be specified in the contract. Quality conformance inpsection for production lots shall be according to MIL-R-23094. The appropriate test level & accept-reject criteria shall be as specified in the applicable specification sheets.

3.4.9 Identification - The following information shall be legibly and permanently marked on the amplifier case or on a nameplate securely attached thereto:

U. S. Navy - BuWeps
Amplifier - Resolver
(Type Designation)
(Specification Sheet No.)
Stock No. (Gov't.)
Manufacturer's Part No.
Contract or Order No.
Manufacturer's Name or Trademark

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3.4.10 Workmanship - Workmanship shall be of the level associated with high grade avionics equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shallbe kept complete and available to the Government as specified, in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of amplifiers shall be classified as follows:

(a) Preproduction tests.
(b) Quality conformance inspection.

4.2.1 Preproduction tests - The preproduction tests of the amplifier shall consist of the tests listed in Table I.

TABLE I
Preproduction Tests
(Tested in the Order Listed)

Rough handling	(4.3.1)
Visual inspection	(4.3.2)
Weight	(4.3.3)
High transient signal test	(4.3.4)
Maximum DC Supply current	(4.3.5)
Power input	(4.3.6)
Phase shift	(4.3.7)
Input impedance	(4.3.8)
Output impedance	(4.3.9)
Internal noise	(4.3.11)
Warm up	(4.3.12)
Voltage gain and linearity	(4.3.13)
Vibration	(4.3.14)
Shock	(4.3.15)
Humidity	(4.3.16)
Salt spray	(4.3.17)
Sand and dust	(4.3.18)
Fungus	(4.3.19)
Acceleration	(4.3.20)
Radio interference	(4.3.21)
Reliability	(4.3.22)

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4.2.1.1 Preproduction test sample - The number of the preproduction test sample shall be as specified in the contract or purchase order. It shall consist of resolver amplifiers manufactured and packaged in accordance with this specification. The units submitted for preproduction testing shall not have been previously tested except for the Individual tests. The samples shall be forwarded to the testing laboratory designated by the procuring activity.

4.2.1.2 Preproduction test sample identification - The test samples shall be plainly identified by durable tags, securely attached, and marked with the information listed below:

Sample for Preproduction Testing

Amplifier, Resolver (Type as designated in the specification sheets)

Submitted by (Manufacturer's name, date) for Preproduction Testing in accordance with MIL-A- under contract or order number (Contract or Order No.)

Manufacturer's Part Number

4.2.2 Quality conformance inspection - The quality conformance inspection of the amplifiers shall consist of Individual tests and Sampling tests.

4.2.2.1 Individual tests - Each amplifier shall be subjected to the tests listed in Table II.

TABLE II

Individual Tests
(Perform Tests in Order Listed)

	Visual inspection	(4.3.2)
	Phase shift	(4.3.7)
	Input impedance	(4.3.8)
	Output impedance	(4.3.9)
*	Voltage gain and linearity	(4.3.13)

* Open loop only

4.2.2.2 Sampling tests - Samples of the product which have passed the Individual tests shall be selected at random in accordance with the following schedule and subjected to the tests listed in Table III.

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Sampling Schedule

Quantity of Amplifiers Offered for Acceptance	Number of Samples
First 15	1
Next 25	1
Next 50	1
Next 100	1
Each additional 200 of fraction thereof	1

Failure of any sample shall be cause for the rjection of the quantity offered for acceptance.

TABLE III

Sampling Tests
(Perform Tests in the Order Listed)

	Weight	(4.3.3)
	High transient signal test	(4.3.4)
	Internal phase shift	(4.3.7)
	Input impedance	(4.3.8)
	Temperature and altitude	(4.3.10)
	Vibration	(4.3.14)
	Shock	(4.3.15)
	Humidity	(4.3.16)
	Salt spray	(4.3.17)
	Sand and dust	(4.3.18)
	Fungus	(4.3.19)
	Acceleration	(4.3.20)
*	Reliability	(4.2.22)

* The samples subjected to the reliability test for determination of the actual MTBF, shall not be a part of the order.

4.2.3 Test conditions - Unless otherwise specified herein or in the applicable specification sheets, all tests shall be performed under the following conditions:

Temperature - Room ambient 25°C ± 5°C
Pressure - Ambient atmospheric
Humidity - Room ambient, up to 90 percent relative humidity

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4.3 Test Methods -

NOTES: (a) Whenever a part of the resolver is used as a component of the amplifier circuit, or a dummy is used as the load, the effect of that part or the dummy on the parameters obtained in the tests shall be taken into consideration. For Example:

In some resolver amplifiers, the resolver primary is center tapped to be used as the dc supply to the transistors of the output stage of the amplifier. In this case, the resolver primary (center tapped) impedance is in parallel with the amplifier output impedance, when the "Output impedance test" is performed.

(b) The summing circuit and the loads in the schematic test diagrams, shall be in accordance with the applicable specification sheets.

4.3.1 Rough handling test - Amplifiers submitted for preproduction tests, shall be packaged and packed in a manner typical of the amplifiers to be delivered under contract or order. Rough handling tests of packaging and packing shall be perfomred on the unopened packages in accordance with MIL-P-7936.

4.3.2 Visual inspection - The amplifier shall be visually inspected to determine conformance to this specification and the applicable specification sheets. Particular attention shall be given to the requirements concerning dimensions affecting interchangeability,materials, marking, finish, and workmanship.

4.3,3 Weight test - The amplifier shall be weighed on a balance of sufficient precision to obtain a value within one percent of the true weight. The weight of the amplifier shall not exceed by more than five percent, the value specified in the applicable specification sheets.

4.3.4 High Transient signal test - The high transient signal test shall be performed as follows:

(a) Connect the resolver amplifier as shown in Figure 1.
(b) Adjust the signal generator to the voltage and frequency specified in the applicable specification sheets.
(c) Apply a step voltage from the value in (b) to the value specified in the applicable specification sheets, hold for one second, and then drop back to the value in (b).

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(d) Repeat the test at the temperature specified in the applicable specification sheets.

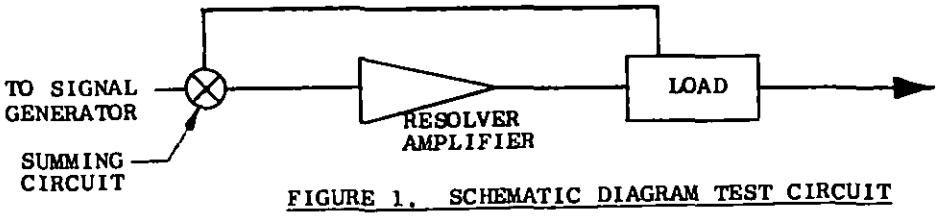


FIGURE 1. SCHEMATIC DIAGRAM TEST CIRCUIT

4.3.5 Maximum DC Supply current - The maximum DC Supply current shall be performed as follows:

(a) Connect the resolver amplifier as shown in Figure 2.
(b) Adjust the signal generator to the voltage and frequency as specified in the applicable specification sheets.
(c) Slowly vary the generator signal from the value in part (b) to the value specified in the applicable specification sheets.
(d) Measure the maximum DC power supply current within the input signal range in part (c).

4.3.6 Power input test - The power input test shall be performed as follows:

(a) Connect the resolver amplifier (closed loop) as shown in Figure 2.
(b) Adjust the dc power supply as specified in the applicable specification sheets.
(c) Adjust the signal generator to the voltage and frequency specified in the applicable specification sheets.
(d) Read the dc power supply current, and calculate the power input to the amplifier.

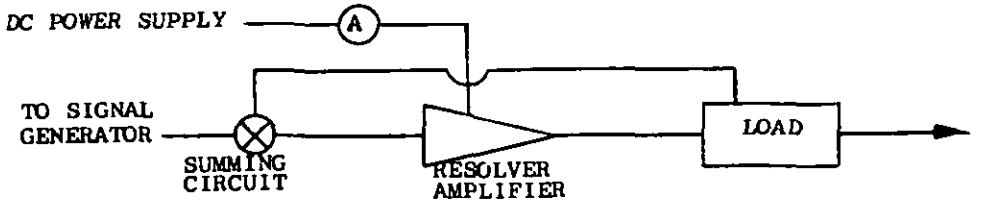


FIGURE 2. SCHEMATIC DIAGRAM TEST CIRCUIT

4.3.7 Phase shift test - The phase shift test shall be performed as follows:

(a) Connect the resolver amplifier as shown in Figure 3.
(b) Adjust the signal generator to the voltage and frequency to obtain at the amplifier output the value specified in the applicable specification sheets.

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(c) Measure the phase shift of the output with respect to the input.

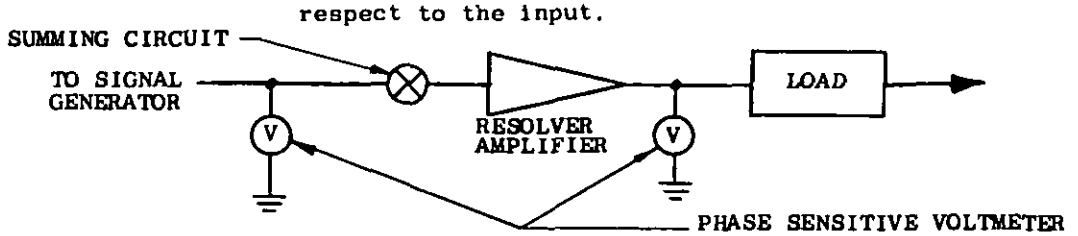


FIGURE 3. SCHEMATIC DIAGRAM TEST CIRCUIT

4.3.8 Input impedance test - The input impedance test shall be performed as follows:

(a) Connect the resolver amplifier as shown in Figure 4.
(b) Adjust the signal generator to the voltage and frequency to obtain at the amplifier output the value specified in the applicable specification sheets.
(c) Adjust Rv to the value which will not cause any variation in the output voltage when the momentary push-button is operated from "ON" to "OFF" position.
(d) Measure the corresponding Rv value. This value is the amplifier input impedance.

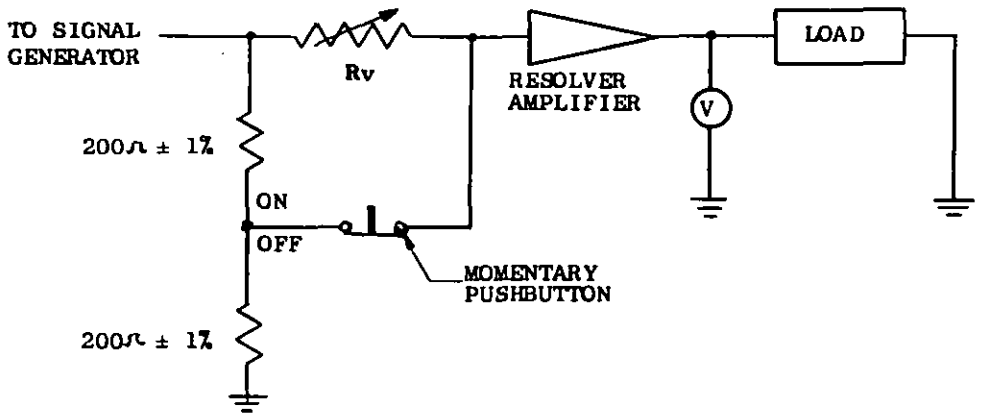


FIGURE 4. SCHEMATIC DIAGRAM TEST CIRCUIT

4.3.9 Output impedance test - The output impedance test shall be performed as specified in the applicable specification sheets.

4.3.10 Temperature and altitude tests - The amplifier shall be subjected to the temperature and altitude tests of the class specified in the applicable specification sheets and MIL-T-5422 with the following exceptions:

(a) At the end of Step 6, the internal noise test (4.3.11) shall be performed.

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(b) At the end of each of the Steps 3, 6, and 14, the voltage gain and linearity tests (4.3.13 only the parts specified in the applicable specification sheets) shall be performed.

4.3.11 Internal noise test - The internal noise test shall be performed as follows:

(a) Connect the amplifier as shown in Figure 5. The resolver shall be in maximum coupling condition.
(b) Measure the output voltage.

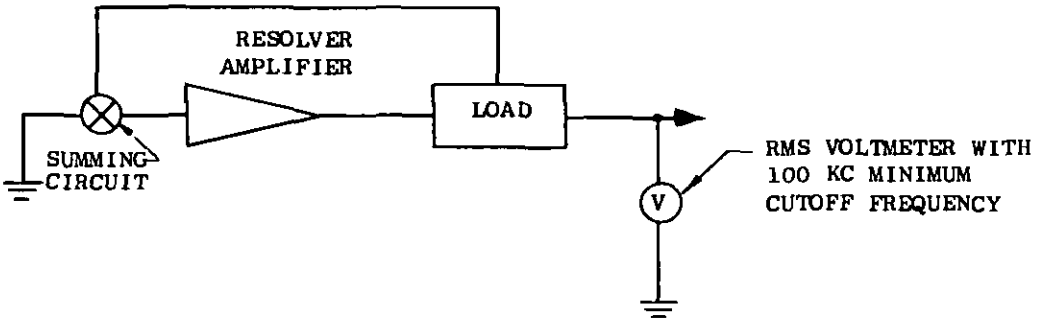


FIGURE 5. SCHEMATIC DIAGRAM TEST CIRCUIT

4.3.12 Warm up test - The warm up test shall be performed as follows:

(a) Connect the resolver amplifier as shown in Figure 6 with the switch in "OFF" position.
(b) Adjust the signal generator to the voltage and frequency as specified in the applicable specification sheets.
(c) Turn the switch to the "ON" position and measure the time for the output signal to settle within 10% of the steady state value.

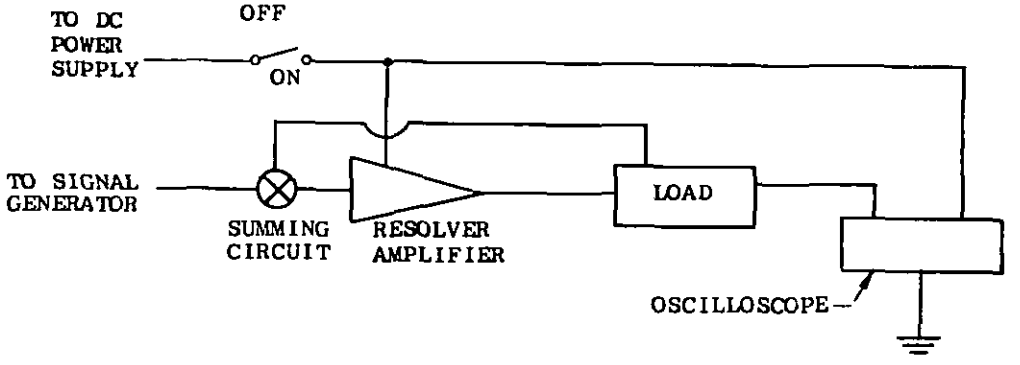


FIGURE 6. SCHEMATIC DIAGRAM TEST CIRCUIT

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4.3.13 Voltage gain and linearity test - The voltage gain and linearity test shall be performed as follows:

Closed loop

(a) Connect the amplifier as shown in Figure 7. Nominal gain.
(b) Adjust the signal generator to the voltage value at 400 ±15 cps as specified in the applicable specification sheets. Measure the output voltage and compute the gain.

Constant voltage versus frequency

(c) Adjust the signal generator frequency down to obtain the lower -3db gain frequency, then up to the lower resonance frequency, then to the upper resonance frequency then to the upper -3db frequency. Record the frequency obtained at each setting.
(d) Scan through the band-width to make sure that no other resonant frequencies are present.

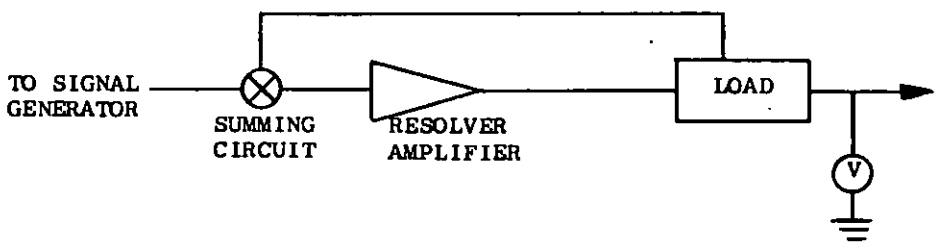


FIGURE 7. SCHEMATIC DIAGRAM TEST CIRCUIT

Maximum input voltage versus hermonic frequencies

(e) Adjust the signal generator frequency to 400±15 cps and to the input voltage specified in the applicable specification sheets.
(f) Vary the input voltage from the value in part (e) up to the maximum value which will not cause any oscillation.
(g) Record the maximum input voltage obtained in part (f).

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(h) Repeat parts (f) and (g) for the frequencies specified in the applicable specification sheets.

Open loop

(i) Connect the amplifier as shown in Figure 8.
(j) Perform the open loop test as specified in the applicable specification sheets.

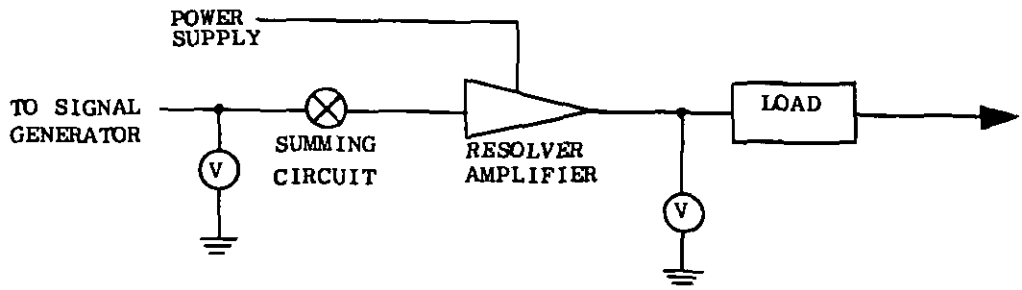


FIGURE 8. SCHEMATIC DIAGRAM TEST CIRCUIT

4.3.14 Vibration test - The amplifier shall be subjected to the vibration test in accordance with MIL-T-5422. Performance check shall be specified in the applicable specification sheets.

4.3.15 Shock test - The amplifier shall be subjected to the shock test in accordance with MIL-T-5422. Performance operation shall be specified in the applicable specification sheets.

4.3.16 Humidity test - The amplifier shall be subjected to the humidity test in accordance with MIL-T-5422 followed by the test specified in the applicable specification sheets.

4.3.17 Salt spray test - The amplifier shall be subjected to the salt spray test in accordance with MIL-T-5422. Performance operation test shall be in accordance with the applicable specification sheets.

4.3.18 Sand and dust test - The amplifier shall be subjected to the sand and dust test in accordance with MIL-T-5422. Performance operation test shall be in accordance with the applicable specification sheets.

4.3.19 Fungus test - The amplifier shall be subjected to the fungus test in accordance with MIL-T-5422. Performance operation test shall be in accordance with the applicable specification sheets.

4.3.20 Acceleration test - The acceleration test shall be in accordance with procedure I of MIL-E-5272. The functional tests shall be in accordance with the applicable specification sheets.

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4.3.21 Radio interference test - Radiated and conducted radio frequency noise shall be within the limits specified in MIL-I-16910 when tested according to the applicable procedure specified therein.

4.3.22 Reliability test - The reliability test shall be performed as follows:

(a) Connect the resolver amplifier as shown in Figure 7.
(b) The resolver amplifier shall be tested in accordance with MIL-R-22973. The test cycle shall be in accordance with the applicable specification sheets. This cycle shall be repeated until the completion of at least the number of hours specified in the contract or order. Time counted toward the reliability test shall be the sum of only the power "ON" periods.
(c) At the end of each four cycles, with the power "ON", the output voltage shall be measured.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Unit packaging - Unless otherwise specified by contract or order, each amplifier shall be individually packaged in a hermetically sealed container in accordance with MIL-P-794 with the cushioning necessary to provide a tight package. The cushioning material shall have a pH value between 6.0 and 8.0, and shall not produce corrosive or deleterious fumes or vapors which could act on or attack the amplifier.

5.1.2 Hardware - All hardware required by the detail specification such as, terminal lugs, terminal screws, etc. shall be packaged in a transparent bag conforming to MIL-B-22205, Type III. The hardware shall be included in the container described in 5.1.1.

5.2 Packing - Unless otherwise specified by contract or order, amplifiers shall be packed in accordance with MIL-P-794, Level A, B, or C as specified by the procuring activity.

5.3 Marking - The nomenclature for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The amplifiers covered by this specification are used as booster amplifiers to isolate, standardize and compensate the computing resolvers found in integrated instrumentation systems, navigational equipments, fire control systems, and control surface systems in shipboard, airborne, and missile applications.

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, date, and number of this specification.
(b) Slant number of applicable specification sheets.
(c) Nomenclature of item.
(d) Level of packaging.
(e) Number of units to be used in the reliability test.
(f) Number of testing hours for the preproduction reliability test.
(g) Number of failures allowed in the preproduction reliability test.
(h) Inspection records of the examination and tests.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-81258(WP) AMPLIFIERS, RESOLVER, GENERAL SPECIFICATION FOR

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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